CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech. (CL) Examination

November 2013

CL 406 Design of Structures - II

Date: 28.11.2013, Thursday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. IS 456:2000, IS 800:2007, IS 875:1987, SP -16, Steel Table are allowed.

SECTION - I

Q - 1 (a) Design a continuous rectangular beam of spans 6 m to carry a dead load of 10 kN/m and a live load of 14 kN/m. The beam is continuous over more than 3 spans and is supported by columns. Take M 20 grade concrete and Fe 415 steel. Sketch the details of reinforcements for the beam. [07]

Q - 2 (a) Design the longitudinal steel for a braced column 500 × 400 mm bent in single curvature with the following data: [07]

Effective length about both axes = 6 m, unsupported length = 6.5 m, factored load = 2000 kN, factored moments on major axis are 240 kNm at the top and 190 kNm at bottom. The factred moments about minor axis are 110 kNm at top and 90 kNm at the bottom. Take M 40 grade concrete and Fe 415 steel.

(b) Design a trapezoidal footing for the two columns A and B transmitting service loads of 750 kN and 1500 kN respectively. The column A is 400 mm square and column B is 600 mm square in size and they are spaced at 4.5 m centers. The property line is 300 mm beyond the face of column A. The safe bearing capacity of soil at site is 140 kN/m^2. Use M 25 grade concrete and Fe415 steel. [07]

OR

Q - 2 (a) Determine the reinforcement required for a braced column against side sway with the following data: [07]

Size of the column = 300 x 400 mm, effective lengths l_{ex} and l_{ey} = 6.0 and 5.0 m, respectively; unsupported length l = 7 m, factored load P_u = 1400 kN, factored moments in the direction of larger dimension = 60 kNm at top and 25 kNm at bottom, factored moments in the direction of shorter dimension = 50 kNm at top and 25 kNm at bottom. The column is bent in double curvature. Reinforcement will be distributed equally on four sides. Take M 30 grade concrete and Fe 415 steel.

(b) Design a rectangular combined footing connecting two columns A and B, 3.5 m centre to centre, carrying an ultimate axial load of 900 kN and 1300 kN respectively. The boundary line of the property is 400 mm from the outer face of the column A. Column A is 360 mm × 360 mm and column B is 420 mm × 420 mm. The bearing capacity of the soil obtained from plate load test is 100 kN/m^2. Use M25 grade concrete and Fe415 steel. Assume effective cover equal to 60 mm. **[07]**

Q - 3 (a) Design a reinforced concrete cantilever retaining wall to retain earth level with the top of wall to a height of 4.5 m above ground level. The density of soil at site is 15 kN/m^3 with a safe bearing capacity of 140 kN/m^2. Assume the angle of internal friction Ø = 30°. The coefficient of friction between soil and concrete as 0.55. Adopt M 25 grade concrete and Fe 415 HYSD bars. Also show reinforcement detail in retaining wall. **[07]**

(b) A longitudinal type of staircase spans a distance of 3.5 m centre-to-centre of beams. The rise R = 175 mm, going G = 250 mm and tread T = 270 mm. The treads have 15 mm granolithic finish and consist of 15 steps. Assuming grade of concrete as M20 and grade of steel as Fe415, design the staircase for a live load of 4.5kN/m^2. Assume breadth of staircase as 1.5 m. **[07]**

OR

Q - 3 (a) Design a cantilever retaining wall to retain earth 4 m above ground level. The density of soil is 17 kN/m^3 and its angle of repose is 30°. The embankment is horizontal at top. The safe bearing capacity of soil is 190 kN/m^2 and the coefficient of friction between soil and concrete is 0.5. Surcharge pressure is 17 kN/m^2. Adopt M 25 grade concrete and Fe 415 HYSD bars. Also show reinforcement detail in retaining wall. **[07]**

(b) A staircase slab spanning longitudinally is set into pockets of two supporting beams, one of either end, in successive landings between the floors. If the effective span of the slab is 3.5 m horizontally and the rise of the stair is 1.5 m with 250 mm treads and 150 mm risers, design and detail the staircase slab. Assume a live load of 3.5 kN/m^2, finishes 0.5 kN/m^2, f_{ck} = 25 MPa and f_y = 415 MPa. **[07]**

SECTION – II

Q - 4 (a) A column section ISHB 400 @ 759.3 N/m carries an axial compressive factored load of 1650 kN. Design a suitable bolted gusset base. The base rests on M15 grade concrete pedestal. Use Fe 410 grade of steel and 24 mm diameter bolts of grade 4.6 for making the connections. **[07]**

Q - 5 (a) Calculate dead load, live load and wind load per panel point for the roof truss shown in **Figure 1** situated in the vicinity of Bombay. **[05]**

Spacing of truss = 3.5 m, Span of truss = 9 m, Slope of truss = 30°,

Height of eaves = 5.5 m, Roofing material = G.I. Sheets, $k_1 = 1$, $k_2 = 0.981$, $k_3 = 1$

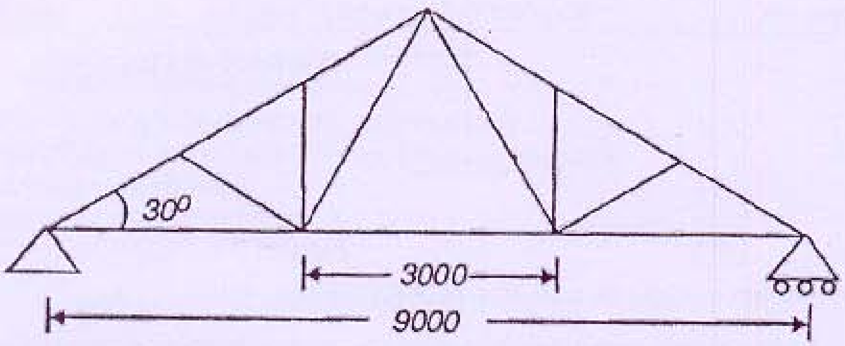

Figure: 1

(b) A gantry girder is to be designed for the following data: **[09]**

Crane Capacity – 100 kN

Centre to centre distance between columns (span of gantry girder) – 8 m

Centre to centre distance between gantry rails – 15 m

Wheel base – 3.2 m, Edge distance – 1.0 m, Weight of crab car – 16 kN

Weight of crane girder – 100 kN

Self-weight of rail section – 300 N/m

Diameter of crane wheels – 150 mm

Design the gantry girder for vertical loads.

<u>OR</u>

(b) A gantry girder section is shown in **Figure 2**. The girder is subjected to following moments: **[09]**

Moment about major axis, $M_z = 600$ kNm

Moment about minor axis, $M_y = 25$ kNm

Check the adequacy of the girder for combined local capacity.

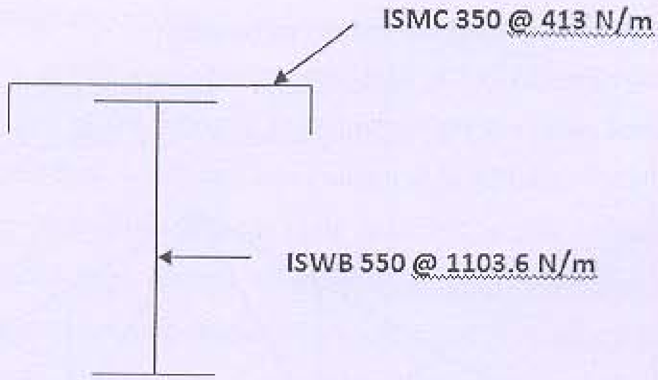

Figure: 2

Q - 6 (a) Design a purlin on a sloping roof truss for the following data: [05]

Spacing of truss = 3.5 m, Spacing of purlins = 1.73 m, Angle of roof truss = 30°

Dead Load = 0.4 kN/m

Live Load = 0.5 kN/m

Wind Load (Suction) = 1.3 kN/m

Use Angle section as a purlin. Do the check for moment capacity.

(b) A plate girder consists of top flanges, bottom flanges and web of sizes 290 x 22 mm, 215 x 22 mm and 900 x 8 mm respectively. Check whether stiffeners are required. If required, suggest the size of vertical stiffeners. [09]

<u>OR</u>

(b) A welded plate girder 20 m in span and laterally restrained throughout. It has to support a uniform load of 110 kN/m throughout the span exclusive of self-weight. Design the girder without intermediate transverse stiffeners. The steel for the flanges and web plates is of grade Fe 410. Design the cross section and do the checks for bending strength. [09]

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